

Extending Equilibrium Matching

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Equilibrium Matching

Problem Statement

Current SOTA generative models (Diffusion, Flow Matching) rely on **non-equilibrium dynamics** (time-dependent vector fields). This enforces fixed integration horizons and prevents flexible optimization-based sampling.

Research Goal

Develop extensions to Equilibrium Matching that enhance inference speed, training efficiency, and stability while rigorously preserving its inherent time-invariant vector field properties.

Tasks

- ▶ Extend EqM to conditional generation and latent spaces.
- ▶ Integrate Second-Order (Hessian) dynamics.
- ▶ Analyze the stability and geometric properties of EqM.

Baseline: Equilibrium Matching

We define a corruption process via interpolation $\gamma \sim U[0, 1]$:

$$\mathbf{x}_\gamma = \gamma \mathbf{x} + (1 - \gamma) \boldsymbol{\epsilon}$$

Unlike Flow Matching, EqM minimizes the difference between the model output $f(\mathbf{x}_\gamma)$ and a target gradient field pointing from noise to data:

$$\mathcal{L}_{\text{EqM}} = \mathbb{E}_{\mathbf{x}, \boldsymbol{\epsilon}, \gamma} \left[\left\| f(\mathbf{x}_\gamma) - \underbrace{(\boldsymbol{\epsilon} - \mathbf{x})c(\gamma)}_{\text{Target Gradient}} \right\|^2 \right]$$

- ▶ **Implicit Energy:** $f(\mathbf{x}) \approx \nabla_{\mathbf{x}} E(\mathbf{x})$.
- ▶ **Optimization Sampling:** Sampling is performed via Gradient Descent (GD) rather than ODE integration.

Main Theorem: Geometry of the Optimal Field

Let $\mathcal{M} \subset \mathbb{R}^D$ be the data manifold and u a corrupted point with distance $r(u) = \text{dist}(u, \mathcal{M})$ and outward unit normal $n(u)$. Under mild concentration and integrability assumptions on the posterior $p(x, \gamma \mid u)$:

Theorem (Asymptotic structure)

As $r(u) \rightarrow 0$, the optimal EqM field decomposes as

$$f^*(u) = A(u) r(u) n(u) + o(r(u)), \quad A(u) := \mathbb{E} \left[\frac{c(\gamma)}{1 - \gamma} \mid u \right].$$

If $|A(u)| \gtrsim 1$, then $\|f^*(u)\| = \Theta(r(u) |A(u)|)$.

What it means. Near the data manifold the field is:

- ▶ **Radial:** it points orthogonally to $\mathcal{M}$ (along $n(u)$);
- ▶ **Proportional to distance:** its strength scales linearly with the distance $r(u)$ to the manifold;
- ▶ **Determined by gain-to-noise ratio:** the coefficient $A(u)$ measures how aggressively the gain $c(\gamma)$ pushes relative to the noise level $1 - \gamma$.

Power-Law Gains and the Vanishing Property

Corollary (Power-law scaling)

For $c(\gamma) = (1 - \gamma)^q$ with $q > 0$ (so $c(1) = 0$),

$$\|f^*(u)\| = \Theta(r(u)^q).$$

Consequently, f^* is bounded near $\mathcal{M}$ and vanishes exactly on it: $f^*|_{\mathcal{M}} \equiv 0$.

Interpretation. The condition $c(1) = 0$ acts as a built-in “brake”:

- ▶ Without it, the expected score would blow up as $r \rightarrow 0$.
- ▶ The factor $(1 - \gamma)^{q-1}$ decays as r^{q-1} , exactly canceling the geometric r growth and leaving a net r^q field.
- ▶ This guarantees **stable optimization**: gradient descent does not oscillate or diverge when it reaches clean data.

Take-away. EqM fields are *manifold-projectors*: they push noise orthogonally onto $\mathcal{M}$ and gently stop when the data is clean.

Future Research Directions

1. **Normalizing Flows and Denoising.** Investigate whether continuous normalizing flows possess an intrinsic denoising property or require explicit geometric constraints to act as manifold projectors.
2. **Adversarial Robustness.** Leverage the projector structure $f^*(u) \approx A(u) r(u) n(u)$ to prove certified defense bounds: show that gradient descent under f^* maps adversarially perturbed points back into a guaranteed tubular neighborhood of $\mathcal{M}$.
3. **Flat vs. Sharp Minima in Generation.** Test the hypothesis that flat energy minima (enforced by Hessian regularization) yield blurred images, whereas sharp minima preserve high-frequency detail. If confirmed, develop adaptive curvature scheduling to balance stability and perceptual quality.